Regulatory data sheet (RDS) – Pharmaceutical market
GELUCIRE 48/16 PASTILLES / GELUCIRE 48/16 PELLETS (Code: 3426)

Last update: 06 November 2023

1. DEFINITION

DEFINITION
Macrogol-32 stearate (type I) EP Polyoxyl-32 stearate (type I) NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Suggested PEG -32 stearate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	33GX5WQC0M (PEG-32 Stearate)
ADDITIONAL INFORMATION ON THE DENOMINATION	Preferred Substance Name: PEG-32 stearate Other names: Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxyoctadecanoate, Polyoxyl 32 Palmitostearate, Polyoxyl Stearate
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (Oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between PEG-32 and Stearic acid 50

COMPOSITION	
Well defined multi-constituent substance composed of PEG-32 mono- and di- esters of Palmitostearic acids (C16 and C18).	
Full Labeling (Key letters system) **A_1 ≥75.0 ; 50.0 ≤ A_2 < 75.0 ; 25.0 ≤ B < 50.0 ; 10.0 ≤ C < 25.0 ; 5.0 ≤ D < 10.0 ; 1.0 ≤ E < 5.0 ; 0.1 ≤ F < 1.0 ; G < 0.1**	
Suggested PEG -32 stearate	A_1 ≥ 75.0%
Other constituents (i.e. incidental ingredients)	
Antioxidant agent: Ascorbic palmitate 100 ppm and Tocopherol 200 ppm	

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4. DRUG MASTER FILE

A Drug Master File (DMF) is a submission to the Food and Drug Administration (FDA) that may be used to provide confidential detailed information about facilities, processes, or articles used in the manufacturing, processing, packaging, and storing of one or more human drugs. The submission of a DMF is not required by law or FDA regulation. A DMF is submitted solely at the discretion of the holder.
(Source: US FDA)

In general, the excipient DMF is not used for compendial excipients but more for novel excipients (e.g., new excipients, mixtures, coprocessed materials, and biotech excipients).
FDA usually does not review DMFs for compendial excipients unless there is a specific reason to do so (e.g., new route of administration or application for existing excipients).
(Source: IPEC Americas)

We have no DMF submitted to FDA for GELUCIRE 48/16 because it is a compendial excipient.

CHINA BUNDLING REVIEW			
(Generic name) Chinese name	**Registration N°**	**Company**	**CFDA/CDE Reference (http://www.cde.org.cn/yfb.do?method=main)**
(PEG-32 Stearate) 聚氧乙烯（32）硬脂酸酯	F20230000203	Gattefossé SAS	Raw material has not been approved yet for use in marketed formulations in Joint-Review with drug applications (Status I)
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	**Conforms to monograph**
European Pharmacopoeia (Ph.Eur.) Current edition	Macrogol stearate (type I)
USP/ NF Current edition	Polyoxyl stearate (type I)

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6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓	✓ (2) (No MRL)	✓	✓ (6)	✓ Polyoxyethylene Stearates (Mol Wt 600-2000)	✓ Polyoxyethylene Stearates (Mol Wt 600-2000) 21CFR180.910 21CFR190.930 Uses: Emulsifier

EUROPEAN REGULATION
(1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption)
(2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units)
(3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether)
(4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin
USA REGULATION
- Veterinary drugs
FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety
(5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR)
(6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database)
- Pesticides
(8)EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)

EUROPEAN REGULATION NO. 470/2009 LAYING DOWN COMMUNITY PROCEDURES FOR THE ESTABLISHMENT OF RESIDUE LIMITS OF PHARMACOLOGICALLY ACTIVE SUBSTANCES IN FOODSTUFFS OF ANIMAL ORIGIN			
REFERENCE	LISTED IN	MRL	ANIMAL SPECIES
Regulation (CE) no 508/1999 Polyethylene glycol stearates with 8-40 oxyethylene units	Annex II (List of substances not subject to maximum residue limits)	Not limited	All species

As detailed in the CVMP publication entitled Position Paper on the definition of substances capable of pharmacological action in the context of Directive 2001/82/EC, as amended, with a particular reference to excipients and manufacturing materials (EMEA/CVMP/072/97-Rev.1), it was concluded that “substances capable of pharmacological action are substances pharmacodynamically active at the dose at which they are administered to the target animal by means of the veterinary medicinal product in which they are included”. It follows that if an excipient has not pharmacodynamic activity at the relevant dose, then no MRL evaluation would be needed.

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7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	**Ingredient Name**	**Use**	**Restrictions**
Approved	PEG-32 stearate Polyethylene glycol 1540 monostearate Polyoxyethylene (32) monostearate	Not available as an Active Ingredient in any application Available for use as an Excipient Ingredient in: Export Only Not available as an Equivalent Ingredient in any applicatio	/
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	**NHPID name**	**Route of administration/ Restrictions**
Approved chemical name	PEG-32 Stearate	Role Non-medicinal: Purposes: Emulsifying Agent Route Of Administration: Topical Detail: Topical use only, up to 10%
Approved chemical name	Polyoxyethylene stearates	Role non-medicinal: Purposes: Emulsifying Agent , Solubilizing Agent , Wetting Agent Detail: Acceptable daily intake up to 25 mg/kg body weight daily as total for polyoxyethylene derivatives
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

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8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Acute irritation (JORF)	Dermal	Rabbit	STEARATE 1500 (PEG-32 stearate) 10% in water	Non Irritant Cutaneous irritation (24h): 0.46	(Gattefossé data, GAT-7904, 1979)
Acute irritation (JORF)	Ocular	Rabbit	STEARATE 1500 (PEG-32 stearate) 10% in water	Non irritant Ocular irritation (48h): 0	(Gattefossé data, GAT-7904, 1979)

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	**Reference**	**Maximum "as used" concentration**	**Conclusions**
PEG Stearates	JACT 2(7): 17-60, 1983	Up to 10% (PEG-32 Stearate)	On the basis of the available information presented in this report, the panel concludes that PEG-2, -6, -8, -12, -20, -32, -40, -50, -100 and –150 Stearates are safe as cosmetic ingredients in the present practices of concentration and use.

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
	CAS #	**EC #**	**Chemical name**
Suggested PEG -32 stearate	9004-99-3	Not applicable (List number : 618-405-1)	Poly(oxy-1,2-ethanediyl), α-(1-oxooctadecyl)-ω-hydroxy-
EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list. EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5. Substances without EC numbers were assigned "list numbers" in order to facilitate REACh submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.			

NATIONAL INVENTORIES					
	U.S.A. (TSCA)	**Canada (DSL/NDSL)**	**Australia (AIIC)**	**China (IECSC)**	**Other inventories**
Suggested PEG -32 stearate	Listed	DSL listed	Listed	Listed	Poly(oxy-1,2-ethanediyl), α-(1-oxooctadecyl)-ω-hydroxy-: AIIC, DSL, ENCS, IECSC, INSQ, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI, AREC, ECL

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NATIONAL INVENTORIES					
	U.S.A. (TSCA)	**Canada (DSL/NDSL)**	**Australia (AIIC)**	**China (IECSC)**	**Other inventories**

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics… are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities.
Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration.
Chemicals not listed on the AIIC must be assessed before importation.

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	**Status per substance**
Suggested PEG -32 stearate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACh regulation.

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10. HISTORY OF USE

10.1. Pharmaceutical applications

Refer to the HISTORY OF USE document.

- No FDA IID reference exactly matches the definition PEG-32 STEARATE
- POLYGLYCERYL STERATE / PEG-8 STEARATE / POLYOXYL 40 STEARATE / POLYOXYL-6 and POLYOXYL-32 PALMITOSTEARATE are the FDA IIG references to the most similar chemical (read-across approach)

10.1.1. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	**Functional Category**	**Regulatory status**	**Safety**
Polyoxyethylene stearates	Emulsifying agent Solubilizing agent Wetting agent	Included in the FDA Inactive Ingredients Database (dental solutions; IV injections; ophthalmic preparations; oral capsules and tablets; otic suspensions; topical creams, emulsions, lotions, ointments, and solutions; and vaginal preparations). Included in nonparenteral medicines licensed in the UK. Included in the Canadian List of Acceptable Non-medicinal Ingredients.	Although polyoxyethylene stearates are primarily used as emulsifying agents in topical pharmaceutical formulations, certain materials, particularly polyoxyl 40 stearate, have also been used in intravenous injections and oral preparations. Polyoxyethylene stearates have been tested extensively for toxicity in animals and are widely used in pharmaceutical formulations and cosmetics. They are generally regarded as essentially nontoxic and non-irritant materials. Polyoxyl 8 stearate LD50 (hamster, oral): 27 g/kg LD50 (rat, oral): 64 g/kg Polyoxyl 20 stearate LD50 (mouse, IP): 0.2 g/kg LD50 (mouse, IV): 0.87 g/kg

COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS		
Denomination	**Category (concentration in %)**	**regulatory status**
Polyoxyl Stearates	Emulsifying agent	1, 4, 10, 12, 16, 23–25
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopeias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portugese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian Garg et al., *Pharmaceutical Technology, Drug Delivery*, 2001, p14-24, Compendium Of Pharmaceutical Excipients For Vaginal Formulations		

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10.2. Food/Nutraceutical/Dietary supplement applications

<table>
<tr><th colspan="4">CONFORMITY TO FOOD ADDITIVE REGULATION</th></tr>
<tr><th colspan="3">THE INGREDIENT IS APPROVED FOR USE IN NUTRACEUTICAL PRODUCTS / DIETARY SUPPLEMENTS IN</th><th rowspan="2">DETAILS ON FOOD ADDITIVE STATUS</th></tr>
<tr><th>EU</th><th>US</th><th>Japan</th></tr>
<tr><td>⦸</td><td>⦸</td><td>⦸</td><td>none</td></tr>
<tr><td colspan="4">FCC: Food chemical codex ; GRAS: Generally recognized as safe ; USFA : US food additive ; USIFA: US indirect food additive ; E xxx: European food additive ; JSFA : Japanese standard food additive ; GB-xxx: Food additive hygien standard (China) ; TFA xxxxx : Taiwan food additive ; KFDA: Korea Food Additives ; DSHEA: USFDA Dietary Supplement Health and Education Act</td></tr>
</table>

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	Not approved
E.U.	EFA (European Food additives)	Not approved
U.S.A.	FCC (Food Chemical Codex)	Not approved
U.S.A.	USFA US Food additives according to 21 CFR	Not approved NB: Glycerides and Polyglycides of hydrogenated vegetable oils (multiconstituant substances containing PEG-stearate) is recognized as Food Additives Permitted For Direct Addition to Food for Human Consumption (21 CFR § 172.736)
U.S.A.	GRAS US Food additives according to 21 CFR	Not approved
JAPAN	JSFA (Japanese Standards of Food Additives)	Not approved

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

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